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into the approval of GM foods—more than a dozen are still in the pipeline—when the approval process is itself beyond reproach. Ramesh says there is “no hurry” and “no overriding food security issues” associated with Bt brinjal. He says, “If long-term toxicity studies have to be done, they should be done.” In effect, it will take about three years and would effectively prevent the introduction of Bt brinjal.

This amounts to the minister admitting that the cart was put before the horse. If the Government is truly open to the idea, if not eager, of introducing GM foods, it should arm itself with independent scientific evidence. “Biotechnology is a powerful tool and always evolving. The risks and benefits should be carefully evaluated. After all, it is the public who has to consume it, not the Government,” points out agriculture scientist and Rajya Sabha member Dr M.S. Swaminathan.

As essential as the independent testing agency is the legal framework that would allow GM companies to be liable for any damage arising from their genetic manipulation. “The Government should put in place a liability provision in the Environment Protection Act which makes a GM crop developer solely liable for any potential leakage and contamination,” says Kavitha Kuruganti of the Kheti Virasat Mission. Creating the legal and testing capability that would address popular concerns over GM foods is, indeed, the only way out of the imbroglio. For this the Government should also put in place a robust framework of protocols that would be independent of the com-

panies which seek approvals. GM foods are normally tested for allergic reactions, gene transfer—genes from the GM foods transferring inside the gut into benign bacteria present there and creating effects such as resistance to antibiotics—and outcrossing in which GM crops get mixed up with normal crops, contaminating them.

Significantly, there is no study on health concerns that shows any adverse effect of the consumption of GM vegetables on humans—in the US, where standards are most stringent, humans consume GM corn and GM soya



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on a massive scale with no side-effects. On the other hand, there is plenty of evidence that shows the harmful effects of consuming the amount of pesticides that end up being used in the cultivation of regular brinjal. Ramesh seems to have given short shrift to the role of pesticides in human consumption. “The first thing that should put us on our guard is the fact that for the first time we are talking of genetically modifying a vegetable that is a common food of nearly daily use in most of our homes. The brinjal is used for human consumption directly and not processed into bread or used in other processed foods,” says Sunita Narain,

Director, Centre for Science and Environment (CSE). She adds, “In many parts of the country, it is even eaten uncooked. We, therefore, need to be extremely cautious in our review of this food. There is a scientific and regulatory black hole.”

Instead of doing justice to all, Ramesh succeeded in raising more serious doubts about the integrity and rationale of the existing institutional structure of the GEAC and the Ministry of Environment to clear GM technology. With the GEAC being dismissed as redundant and a new regulatory law still to be put into practice, there are fresh fears about future tests. “There is no clarity. Transgenic food items do not fall under the ambit of the food safety rules in the country. The moratorium falls short of specifying anything and that is not acceptable,” says organic farmer Vivek Cariappa and promoter of the Krac-a-Dawna

Organic Farm in Karnataka.

Ramesh has also expressed his strict opposition to private participation in the country’s biotechnological research in agriculture. “Biotechnology in agriculture should rest with public sector research institutes,” contends the minister, oblivious to the intense adverse impact in an age when food needs, including its research and food security, are viewed as global challenges. “This decision is a big setback for biotechnology. It will thwart investment in biotech R&D,” says Kiran Mazumdar-Shaw, CMD, Biocon.

Evidently, whether it is Bt brinjal or not, the minister appears to have bitten off more than he can chew. Or maybe muddled policy-making is the price to pay for getting an economist-turned-politician to take hard-nosed decisions in which science can’t be ignored. ■



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